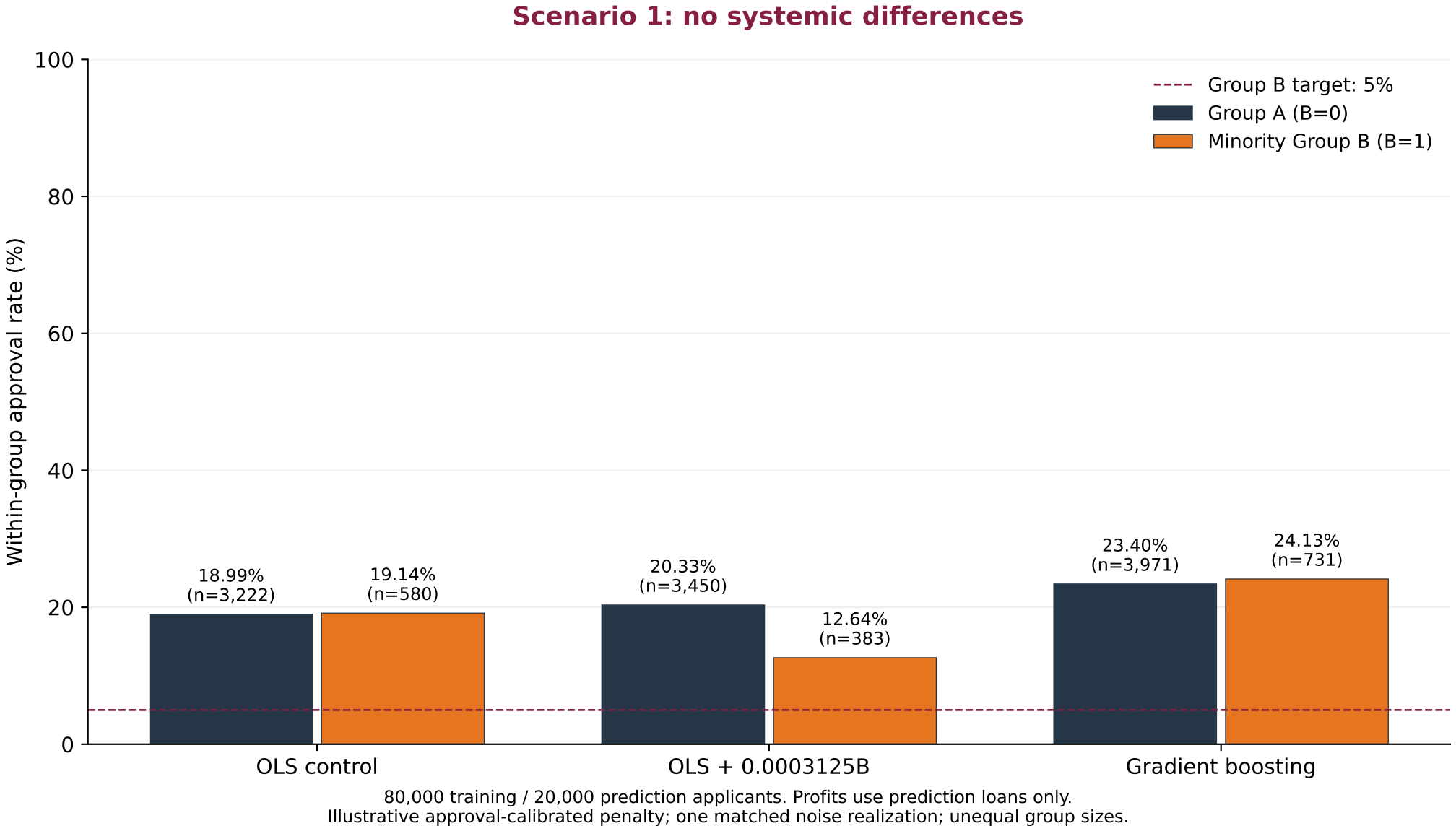
Scenario 1: no systemic differences
Group B target: 5%
Group A (B=0)
Minority Group B (B=1)
Within-group approval rate (%)
100
80
60
40
20
0
18.99%
(n=3,222)
19.14%
(n=580)
20.33%
(n=3,450)
12.64%
(n=383)
23.40%
(n=3,971)
24.13%
(n=731)
OLS control
OLS + 0.0003125B
Gradient boosting
80,000 training / 20,000 prediction applicants. Profits use prediction loans only.
Illustrative approval-calibrated penalty; one matched noise realization; unequal group sizes.